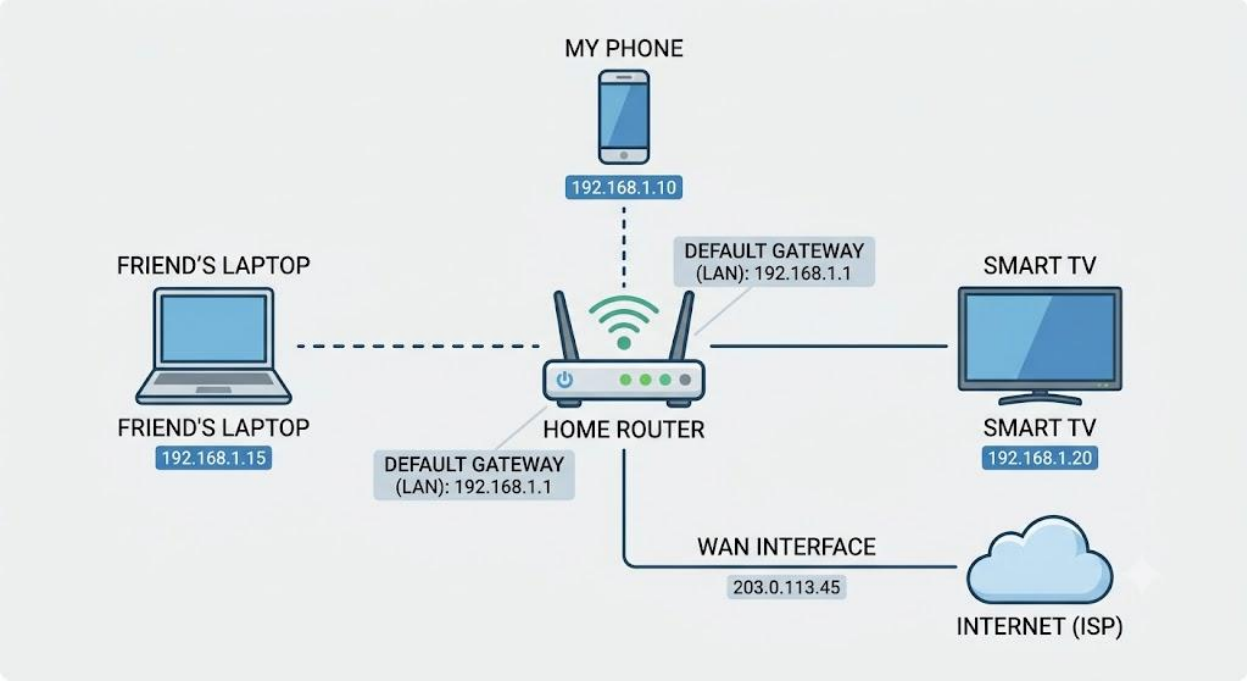
Assignment 9

IP Routing & Routing Protocols

1. **Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.**



2. **Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.**

❖ **Example: Sending a packet to a friend's computer in a different subnet**

1. My laptop IP is 192.168.1.10/24, and my friend's computer IP is 192.168.2.20/24.
2. My laptop checks its routing table and sees that the friend's IP is in a different subnet, so it sends the packet to the default gateway 192.168.1.1.
3. Router 1 checks its routing table and finds that 192.168.2.0/24 is reachable through Router 2.
4. Router 2 checks its routing table, finds the destination network connected to it, and forwards the packet to my friend's computer.
5. The packet reaches 192.168.2.20, and the routing tables help each router decide where to send it next.

3. **Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.**

❖ **Zomato Delivery Network Routing Table**

Branch	Network Address	Subnet Mask	Next Hop	Interface
Ahmedabad	192.168.1.0	255.255.255.0	Direct	GigabitEthernet0/0
Delhi	192.168.2.0	255.255.255.0	10.0.0.2	GigabitEthernet0/1
Mumbai	192.168.3.0	255.255.255.0	10.0.0.6	GigabitEthernet0/2

➢ **Explanation:** Each branch has a different network address. The router checks the destination IP, finds the matching route, and forwards the packet through the specified interface to the next hop. These are fictional example addresses.

4. Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

❖ **Static Routing vs Dynamic Routing**

Routing Type	Pros (Advantages)	Cons (Disadvantages)
Static Routing	1. Easy to set up in a small network. 2. Uses fewer router resources. 3. Gives the admin full control over routes.	1. Routes must be changed manually. 2. Does not automatically find another route if a link fails. 3. Difficult to manage in large networks.
Dynamic Routing (RIP/OSPF)	1. Automatically learns routes. 2. Can find another route if a link fails. 3. Useful for large networks, such as Zomato branches in different cities.	1. Uses more CPU and memory. 2. More complex to configure. 3. Routing updates can use network bandwidth.

- **Real-world example:** A small restaurant can use static routing because its network is simple. A food delivery service like Zomato can benefit from dynamic routing to connect multiple branches and find alternative network paths when a connection fails.

5. **Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.**

❖ **OSPF Configuration Between Two Routers**

Router 1 (R1)

- **enable**
- **configure terminal**
- **router ospf 1**
- **network 192.168.1.0 0.0.0.255 area 0**
- **network 10.0.0.0 0.0.0.3 area 0**

Router 2 (R2)

- **enable**
- **configure terminal**
- **router ospf 1**
- **network 192.168.2.0 0.0.0.255 area 0**
- **network 10.0.0.0 0.0.0.3 area 0**

❖ Explanation of each line

- **enable**: Enters privileged mode.
- **configure t**: Opens configuration mode.
- **router ospf 1**: Enables OSPF routing process 1.
- **network** 192.168.1.0 0.0.0.255 area 0: Advertises Router 1's local network.
- **network** 192.168.2.0 0.0.0.255 area 0: Advertises Router 2's local network.
- **network** 10.0.0.0 0.0.0.3 area 0: Enables OSPF on the shared router-to-router network.
- **area 0**: Places the networks in OSPF's backbone area.